MIL-C-7958A
28 July 1971
SUPERSEDING
MIL-C-7958
18 July 1952

MILITARY SPECIFICATION

CONTROLS, PUSH-PULL, FLEXIBLE AND RIGID

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flexible and rigid push-pull controls for use in activating remotely located equipment.

1.2 Classification. Push-pull controls shall be furnished in the following types, as specified:

Type I	Push-pull controls with flexible tubular casing containing a moveable element.
Type II	Push-pull controls with rigid tubular casing containing a moveable element.
Type III	Push-pull controls with a combination of rigid and flexible tubular casing containing a moveable element.

In addition, push-pull controls shall be designated Grade A or Grade B to indicate precision quality or general utility quality items as defined in Section 6 of this specification.

*2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements For
PPP-B-601	Box, Wood, Cleated-plywood
PPP-B-636	Box, Fiberboard

FSC 2995

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Military

MIL-P-116	Preservation, Methods of
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys

STANDARDS

Federal

FED-STD-151	Metal, Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-143	Standards and Specifications, Order of Precedence, For the Selection of
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing of
MIL-STD-838	Lubrication of Military Equipment
MS-27975	Clevis, Rod End-adjusting, Wide and Narrow Forks
AN-665	Terminal Threaded Clevis Type Tie Rod

HANDBOOKS

MIL-HBK-275	Guide For Selection of Lubricants, Fluids, and Compounds For Use In Flight Vehicles and Components

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.0.1 First article inspection. This specification makes provisions for first article inspection.

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3.1 Materials. Materials shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications or which are not specifically described herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.1.1 Metals. - All metals used in the construction of push-pull controls shall be of a corrosion-resisting type, or shall be suitably protected to resist corrosion during the normal service life. The use of dissimilar metals, especially brass, copper, or steel, in contact with aluminum or aluminum alloy shall be avoided wherever practicable.

3.1.2 Selection of materials. - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.1.2.1 AN or JAN standard parts. - AN or JAN standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the AN or JAN standard parts without alteration, and provided the corresponding AN or JAN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or JAN standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2 Design and construction. - Push-pull controls shall be of the types and grades specified in paragraph 1.2, and shall be capable of transmitting tension and compression forces applied to either end of the moveable element. The moveable element may be either removable or permanently assembled. If removeable, it shall be capable of being quickly and easily removed, without the use of special tools, for the purpose of inspection and relubrication.

3.2.1 Shape. - Control units shall be shaped or formed as specified on the installation drawing or template furnished by the procuring activity.

3.2.2 Tubular casing. - The tubular casing shall be of the types specified in paragraph 1.2. It shall be of metallic construction and may be encased by a suitable nonmetallic covering. The tubing casing may consist of detachable sections for ease in shipping.

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3.2.2.1 Seals. - Seals shall be provided to keep the control unit moisture, dust, and lubrication-tight where necessary.

3.2.2.2 Ends. - Unless otherwise specified, the tubular casing shall have self-aligning ends capable of an angular displacement of at least 10 degrees from the normal position in any plane.

3.2.3 Moveable element. -

3.2.3.1 Terminal ends.- The ends of the moveable element shall be threaded with an external thread, .250-28 UNF-3A, in accordance with MIL-S-7742, to accomodate the MS27975-3 or 4 rod end clevis, the AN665-34R terminal or other internal threaded antifriction or plain bearing self-aligning rod ends.

3.2.3.2 Stroke. - The length of stroke of the moveable element shall be as specified by the procuring activity.

3.2.4 Weight.- The weight of the push-pull control elements shall not exceed the weights shown in table I.

TABLE I
Weight of Controls

Item	Pounds (max)
Control per foot of length with rigid casing	0.30
End fittings for rigid casing (4-inch stroke)	0.47 (total-both ends)
Control per foot of length with flexible casing	0.58
End fittings for flexible casing (4-inch stroke)	0.62 (total - both ends)

3.2.5 Backlash. - When the design of the control is such that lost motion can develop within the moveable member during normal service, a positive adjustment shall be provided in the moveable member capable of absorbing this lost motion.

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3.2.6 Finish. -

3.2.6.1 Aluminum-alloy parts. - Exposed aluminum-alloy parts shall be anodized in accordance with MIL-A-8625.

3.2.6.2 Steel parts. - All exposed steel parts, unless fabricated of corrosion-resisting steel, shall be cadmium-plated in accordance with QQ-P-416, or zinc-plated in accordance with QQ-Z-325.

3.3 Identification of product. -

3.3.1 Each control shall be permanently and legibly marked with the following information at a convenient location:

MIL-C-7958A
Manufacturer's Part No.
Manufacturer's Name or Trade-Mark

3.3.1.1 Manufacturer's part number. - The part number shall be the same as the number assigned to the manufacturer's assembly drawing, plus the applicable dash number if the drawing is tabulated. The part number shall distinctly identify the control from all similar controls which are produced against a basic design but varying in length, distance of travel, markings, etc.

3.4 Performance. -

3.4.1 General. - The control unit as shown in figure 1 shall meet the performance values specified in table II for:

(a) Backlash
(b) Operating Force with no Load
(c) Operating Force under Tension and Compression Loads
(d) Break Away Force

3.4.2 Endurance. - After 25,000 reversals of motion, the operating force, the friction under tension, and compression loads shall not exceed the values shown in table II at room temperature. The change in backlash as a result of this test shall not exceed 0.005 inch.

3.4.3 Lubrication. - For all controls except those permanently assembled during manufacture, lubrication shall be in accordance with MIL-STD-838. The lubricant shall be selected from MIL-H-5606-275, and shall be of such quality that the control shall meet the requirements of table II. The control shall show no leakage of lubricant after operation at a temperature of 88° ±2°C (190° ±4°F) for a period of 30 minutes, and shall operate satisfactorily while at this temperature.

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TABLE II

Performance Values

Test 1/	Room Temperature		70° ±1°C (158° ±2°F)		-55° ±1°C (-67° ± 2°F)	
	Grade A	Grade B	Grade A	Grade B	Grade A	Grade B
Backlash	.020	.026	—	—	—	—
Operating Force with No Load (max)	2.6 lb	4.0	2.6 lb	4.0	15 lb	20
Operating Force with 5-lb, Tension and compression Load (max)	11.1 lb	15.0	11.1 lb	15.0	23 lb	30
Operating Force with 25 lb, Tension and compression Load (max) 2/	52 lb	55	52 lb	55.0	60 lb	65
Breakaway Force with No Load (max)	—	—	—	—	20 lb	25

1/ Rigid controls shall show no visible distortion, especially at the bends, while under these loads. Abnormal distortion shall not be exhibited by the flexible controls while under these loads.

2/ Soaking periods prior to test at the temperatures specified in the table shall be 4 hours minimum.

3.4.4 Corrosion resistance. - After being subjected to salt spray for 50 hours, the operating force of the controls shall not exceed 50-percent (max) increase over the original operating force observed during the room temperature operating force of the particular control.

3.5 Workmanship. - The push-pull control, including all parts shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given fits, retention of lubricant, and freedom of parts from burrs and sharp edges.

*3.6 Approvals for production. -The controls furnished under this specification shall be of a design, model, or type which has satisfactorily passed the first article tests.

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4. QUALITY ASSURANCE PROVISIONS

*4.01 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1 Classification of tests. The inspection and testing of push-pull controls shall be classified as follows:

(a) First article inspection: First article tests are those tests performed on samples to ascertain that the design, model, or type is capable of complying with the specified performance requirements and is suitable for volume production. (See 4.2)

(b) Acceptance tests: Acceptance tests are those tests accomplished on products submitted for acceptance under contract (See 4.3)

*4.2 First article inspection.

4.2.1 Sampling instructions. - Samples for first article tests shall consist of one control of each design, shaped and mounted as shown in figure 1. Samples shall be forwarded to the procuring activity and shall be plainly identified by securely attached durable tags, marked with the following information:

Sample for First Article Inspection
Controls; Push-pull, Flexible or Rigid (as applicable)
Name of Manufacturer
Submitted by (name of manufacturer) (date) for first article testing in accordance with the requirements of MIL-C-7958A under authorization (reference authorizing letter)

Note: Two sample controls will be required if the control is of the permanently assembled type.

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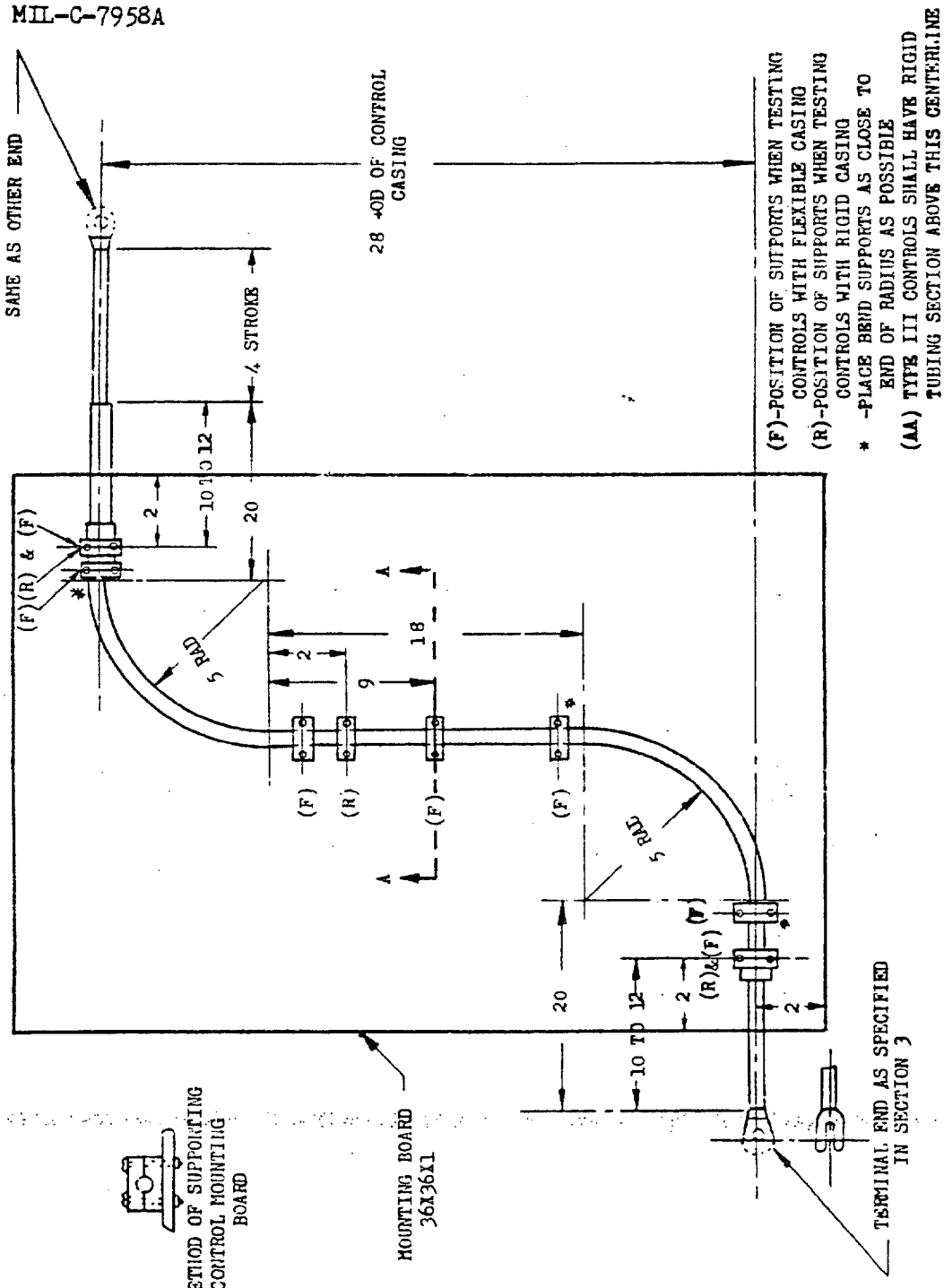


FIGURE 1

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4.2.2 Tests. - The first article tests of push-pull controls shall consist of all the tests of this specification in the following order, as described under Test Methods (see paragraph 4.5)

(a) Examination of Product
(b) Backlash
(c) Operating Force with No Load
(d) Operating Force under Tension and Compression Loads
(e) Humidity (conduct prior to low temperature test)
(f) Lubrication (conduct following high temperature ($70^{\circ} \pm 1^{\circ}C$) test
(g) Break-Away Force
(h) Endurance
(i) Corrosion-Resistance

*4.3 Acceptance tests. - The acceptance tests shall consist of all sampling tests described in paragraph 4.3.1.

4.3.1 Sampling tests. - A random sample shall be taken according to MIL-STD-105 using inspection level IV and AQL Class 1.2 to 2.2, and the rejection provisions contained therein. Each push-pull control in the sample shall be submitted to Examination of Product, as described under Test Methods. If the sample fails this test, the entire lot shall be examined in detail by the contractor, defectives discarded, and the lot resubmitted to the Government Inspector for sampling according to the above procedure; but using inspection level V and AQL Class 0.65 to 1.2. A lot shall consist of all controls of the same type and size submitted for inspection at the same time and place. If this second sample fails to pass, the lot shall be rejected.

4.3.1.1 In addition, controls may be subjected to any of the other tests specified herein which the Inspector considers necessary to determine conformance with the requirements of this specification.

4.3.2 Rejection and retest. - Failure of any sample control to conform to any one of the tests shall be cause for the rejection of the lot represented. Controls which have been rejected may be reworked or replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Controls rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.4 Test conditions. -

4.4.1 Room temperature. - Where room temperature is specified, it shall be from 20° to $29^{\circ}C$ (68° to $85^{\circ}F$).

4.4.2 Backlash adjustment. - No adjustment of the controls shall be permitted during the entire test program.

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4.5 Test methods. -

4.5.1 Examination of product. - The controls shall be carefully examined to determine conformance to the requirements of this specification not otherwise covered by tests.

4.5.2 Backlash. - Backlash shall be determined by securing the output end of the control unit and subjecting the input end to tension and compression loading of 5 pounds. The total travel of the input end shall be recorded as backlash and shall not exceed the limits specified in table II.

4.5.3 Operating force with no load. - The operating force required to push or pull the controls shall be measured by any suitable, accurate means and shall not exceed the limits specified in table II.

4.5.4 Operating force under tension and compression loads.- The operating force shall be measured by any suitable, accurate means and shall not exceed the limits specified in table II. The specified loads shall be applied to the control by any suitable means. The operating force with the control in tension shall be measured by pulling the control. The friction with the control in compression shall be measured by pushing the control.

4.5.5 Humidity. - The equipment shall be placed in the test chamber, and the temperature and relative humidity raised to 70° ± 2°C (158 ± 3.6°F) and 95 ± 5 percent, respectively, and maintained at such for a period of 6 hours. At the end of the 6-hour period, the heat shall be shut off. During the following 16-hour period, the temperature must drop to 38°C (100.4°F), or less. At the end of the 16-hour period, heat shall again be applied for an additional 2-hour period to stabilize to 70° ± 2°C (158° ± 3.6°F). This cycle shall be repeated five times for a total of 120 hours. At the end of the 120-hour period, the control shall meet the requirements of the low temperature provisions of table II.

4.5.5.1 Test chamber. - The test chamber shall be capable of maintaining an internal temperature of 70° ± 2°C (158° ± 3.6°F) and an internal relative humidity of 95 ± 5 percent. The test chamber shall be capable of being sealed so as to retain the total moisture content in the test space. The heat loss from the chamber shall be sufficient to reduce the internal temperature from the above specified operating temperature to not more than 38°C (100.4°F) within a period of 16 hours from the time of removal of the source of heat. Distilled or demineralized water shall be used to obtain the required humidity.

4.5.6 Lubrication. - After attaining the specified temperature, the control shall be operated through its full stroke at the rate of 20 cycles per minute for the specified time. The control shall then be examined for any sign of lubricant leakage.

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4.5.7 Break-away force. - The break-away force shall be measured at the end of the low temperature soaking period specified, and shall be the maximum force measured to pull the controls on the first movement after the soaking period.

4.5.8 Endurance. - The test shall be conducted at room temperature. The reversals shall be applied at the rate of 20 cycles per minute with 4-inch stroke until the specified number of reversals have been completed. The loading shall be such that it imposes a 5-pound compression load on the assembly during the push stroke and a 5-pound tension load on the assembly during the pull stroke.

4.5.9 Corrosion resistance.- The control shall be operated through its full stroke at half-hour intervals for the specified time during the Salt-Spray treatment in accordance with FED-STD-151. Following this test, the controls shall be subjected to the tests headed Backlash, Operating Force, and Friction Under Tension and Compression Loads, at room temperature. The controls shall be disassembled and examined for evidence of wear or distortion. The casing shall be sectioned at the bends and examined for decrease in thickness.

*5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see paragraph 6.2).

5.1.1 Level A. - Each pushpull control shall be preserved and packaged Method IC-2 in accordance with MIL-P-116.

5.1.2 Level C. - Each push-pull control shall be preserved and packaged in a manner to prevent deterioration and damage during shipment from the supplier to the first receiving activity for immediate use.

5.2 Packing. - Item preserved and packaged in accordance with paragraph 5.1 shall be packed level A, B, or C as specified (see paragraph 6.2).

5.2.1 Level A.-Items shall be packed in overseas containers conforming to PPP-B-601. The container shall be surface-treated in accordance with the specification.

5.2.2 Level B.-Items shall be packed in containers conforming to PPP-B-636, class weather-resistant.

5.2.3 Level C. - Push-pull controls preserved and packaged as specified in paragraph 5.1 shall be packed in fiberboard boxes conforming to domestic class of PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. - Push-pull controls covered by this specification are intended for use in aircraft for remote operation of equipment.

*6.2 Ordering data. - Requisitions, contracts, and orders should specify the following information:

(a) Drawing or template for the desired control installation showing length of control, radii and angle of bends, off-sets, length of stroke, and number of self-aligning ends.

(b) Type and grade of push-pull control required.

(c) Whether domestic or overseas packing is desired (see Section 5).

(d) Where the first article test samples should be sent, the activity responsible for testing, and instructions concerning the submittal of the test reports.

6.3 Definitions. -

6.3.1 Backlash. - The term backlash as used in this specification shall be interpreted as follows: the motion lost within the assembly as a result of the clearance between the moveable member and the casing and to include the wear of the individual elements of the moveable member.

6.3.2 Grade A controls. - Grade A controls are push-pull controls made of specially selected components which are individually tested by the contractor to insure that backlash and operating forces are reduced to the minimum values practicable.

6.3.3 Grade B controls. - Grade B controls are push-pull controls made to high-quality commercial standards for applications not requiring Grade A controls.

Note: The margins of this specification are marked with an asterick to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done

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Custodian:

Army - AV
Navy - AS
Air Force - 71

Reviewing Activity:

Navy - AS
User Activity:
Army - AT

Preparing activity:

Air Force - 71

Project Number:
2995-0082

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15413

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